

Data Fairness Reflection

The standard I used to decide whether a neighborhood counts as under-represented:
I made a group by neighborhood by counting all unique values for member id to check the number of members .
seeing the portions of the neighborhoods to each other and how one of them is too low than the other like we will see with Shubra and Maadi .

So we can say that a neighborhood could be counted as under-represented when we see huge gaps like 3 times the average of the other neighborhoods or the highest one .

I also used group by neighborhood with checkout_date count to get all checkout made by the members of these neighborhoods

For both groups I used matplotlib to visualize the results .

My findings:

I found that Shubra was under-represented in the data . Zamalek too but it's with less degree or in other words we can say that it could be a normal graduation to the data not a real one. So I count it a normal graduation or moderately-lower not under-represented. We can see that neighborhoods like Maadi ,Nasr City and Heliopolis had a number of members of 22 , 20 , 18 respectively, on the other hand Shubra and Zamalek were 6 and 14 respectively . We can also see that Zamalek isn't under-represented with a huge gap but Shubra is under-represented with a more than 3 times ratio between it and Maadi.

Results of grouping:

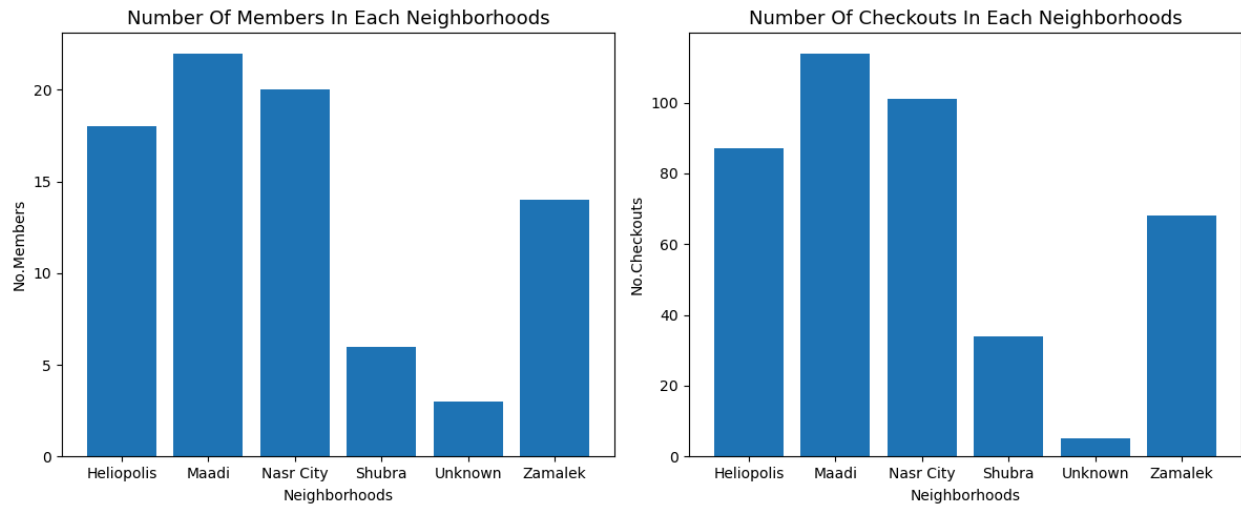
neighborhood	
Heliopolis	18
Maadi	22
Nasr City	20
Shubra	6
Unknown	3
Zamalek	14

Same case in checkouts per neighborhood . I figured that neighborhoods like Maadi , Nasr City and Heliopolis had a number of checkouts of 114 , 101,87 respectively . while Shubra and Zamalek were 68 and 34 respectively . Zamalek still near to the other neighborhoods but Shubra is still too far numerically

Results of grouping:

Heliopolis	87
Maadi	114
Nasr City	101
Shubra	34
Unknown	5
Zamalek	68

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A plausible reason:

Maybe due to that Shubra isn't the center of attraction . Not like Maadi which its residents and the theme of the neighborhood encourage and love reading . Maybe due to less campaigns to neighborhoods like Shubra and putting focus on old classy neighborhoods

A next step

Try to equally reach people by campaigns and events not just Downtown neighborhoods or classy ones